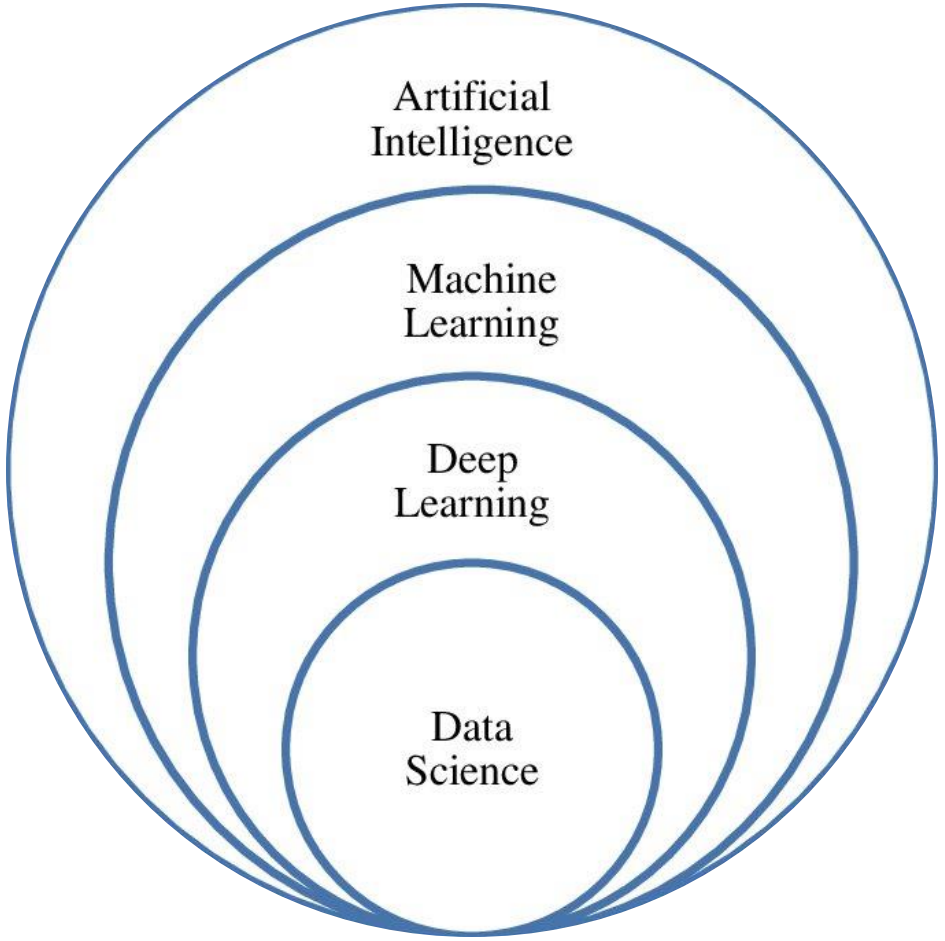
24CSA301-Machine Learning

Lecture: Introduction to ML

What is Machine Learning?

- Machine Learning is a concept which allows the machine to learn from examples and experience, and that too without being explicitly programmed.
- It is the subset of AI.
- *A field of AI that uses statistical techniques and algorithms to enable computer systems to recognize patterns in data, make predictions, classify information, and automate decision-making based on the learned patterns and insights.*



Definition by Tom Mitchell in 1997

- *"A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience"*
- **Task (T):** The specific problem the computer is trying to solve (e.g., classifying emails as spam, predicting housing prices, translating languages).
- **Experience (E):** The historical data provided to the algorithm to learn from.
- **Performance (P):** A quantifiable metric used to evaluate how well the system completes the task (e.g., prediction accuracy or error rate)

Data

- In **Machine Learning (ML)**, **data** refers to the collection of facts, observations, or measurements that are used to **train, test, and evaluate** a machine learning model so that it can learn patterns and make predictions or decisions.

Types of Data

- Based on Structure: Structured, Unstructured and Semi Structured
- Based on Representation: Numerical, Categorical and Ordinal
- Based on Labelling: Labelled and Unlabelled

Based on Structure

Structured Data:

- Highly organized and stored in predefined formats, such as tables, spreadsheets, and relational databases.
- It follows a fixed schema with clear relationships between data points.
- Eg: Customer records in an SQL database, financial transactions in spreadsheets, or inventory management systems.
- Applications: fraud detection, and business intelligence.

Unstructured Data

- lacks a predefined format and does not fit neatly into relational databases.
- It is typically complex, diverse, and requires advanced preprocessing before use in machine learning models.
- Eg: Images, videos, audio files, emails, and social media posts.
- Applications: Sentiment analysis, and speech recognition.

Semi-Structured Data

- Data has some level of organization but does not follow a strict schema like structured data.
- It contains tags or metadata that help define relationships.
- Eg: JSON and XML files
- Applications: Document Classification, web scraping

Based on Representation

- **Numerical (Quantitative) Data**
- data consists of measurable or countable values, making it suitable for statistical analysis and machine learning models. It is further divided into:
- **Discrete Data** – Represents countable values with a finite number of possible outcomes.
  - **Examples:** Number of students in a class, website clicks, number of defects in a product.
- **Continuous Data** – Represents measurable values with an infinite range. It includes valueswith decimal points.
  - **Examples:** Temperature readings, stock prices, blood pressure levels.

Categorical (Qualitative) Data

- Categorical data consists of labels or categories that classify objects or individuals. It is divided into:
- **Nominal Data** – Categories without any inherent order.
  - **Examples:** Gender (Male/Female), Eye color (Brown, Blue, Green), Car brands (Toyota, Ford, BMW).
- **Ordinal Data** – Categories with a meaningful order or ranking, but without a consistent numerical difference.
  - **Examples:** Customer satisfaction levels (Poor, Average, Good, Excellent), Education levels (High School, Bachelor's, Master's, Ph.D.).

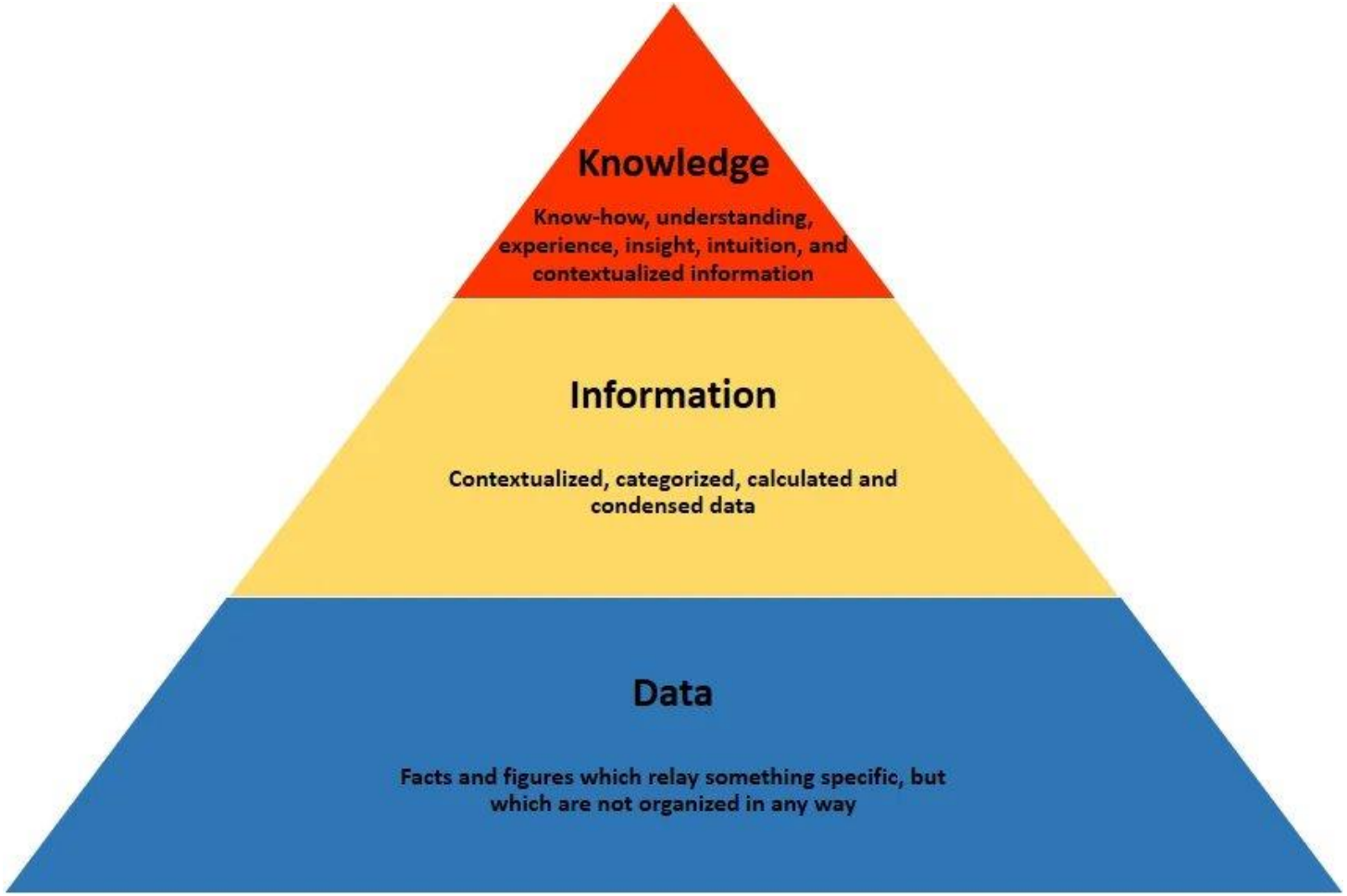
Based on Labelling

- **Labeled Data**
- Labeled data consists of input features paired with corresponding output labels
- This type of data is used for tasks where the relationship between input and output is well-defined.
- **Examples:** Email classification (Spam or Not Spam), Image recognition (Identifying cats vs.dogs).
- **Applications:** Used in classification, regression, and NLP models.

Unlabeled Data

- Data contains only input features without predefined labels, requiring models to identify patterns and relationships on their own.
- **Examples:**
  - Customer segmentation based on purchase behavior.
  - Clustering social media trends.
- **Applications:** Used in clustering, anomaly detection, and dimensionality reduction.

Insightsfrom data



Why ML?

- **Enhancing Decision-Making: Data driven insights, Predictive Analysis**
- **Improving Efficiency and Automation: Automating repetitive tasks (fraud detection)**
- **Personalizing User Experiences: recommendations, marketing**
- **Advancing Healthcare: early disease detection, drug discovery**
- **Enhancing Security: fraud detection**
- **Transforming Industries: autonomous vehicles**
- **Facilitating Scientific Research: data analysis, simulation and modelling**

Characteristics of ML

- Learning from data
- Pattern Recognition
- Generalization
- Adaptive Improvement
- Automation

Goals of ML

- Automate decision making
- Pattern recognition
- Improve accuracy over time/Adapt to changing environment
- Reduce human intervention
- Optimize processes

Applications

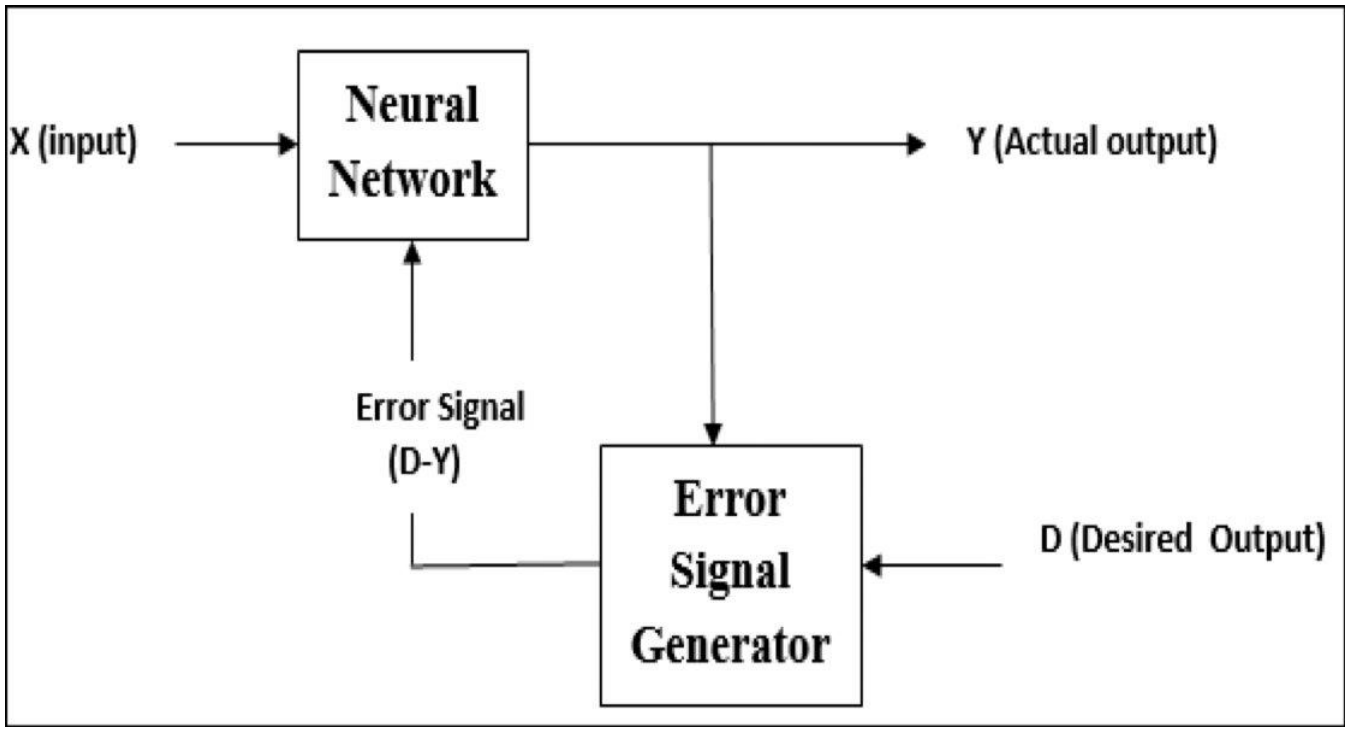
- Healthcare: Disease diagnosis, Drug discovery
- Banking and Finance: Fraud detection, Credit scoring ,Stock market prediction
- E-commerce: Customer segmentation, demand forecasting
- Education: Personalized learning, Adaptive online learning platforms.
- Agriculture: crop disease predictions, soil analysis
- Entertainment: Music/video recommendations
- Natural Language Processing (NLP): Chatbots, speech recognition
- Computer Vision: Face recognition, medical Imaging

Types of learning

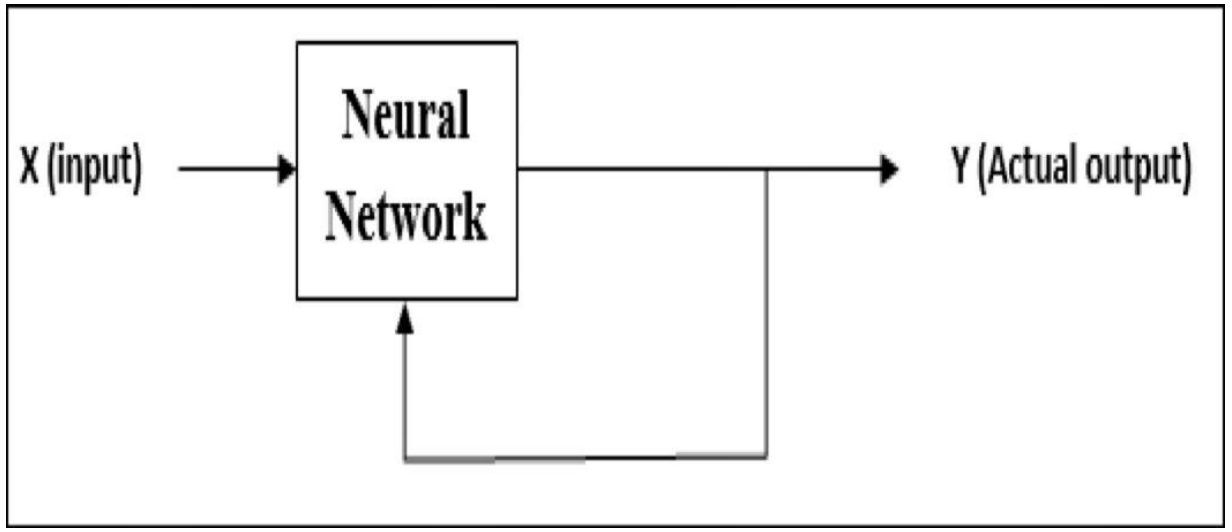
- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Supervised Learning

- The technique that uses labeled data sets to train the models to identify the underlying patterns and relationships.
- It learns the relationship between features and target from the input data.
- Types: Classification and regression



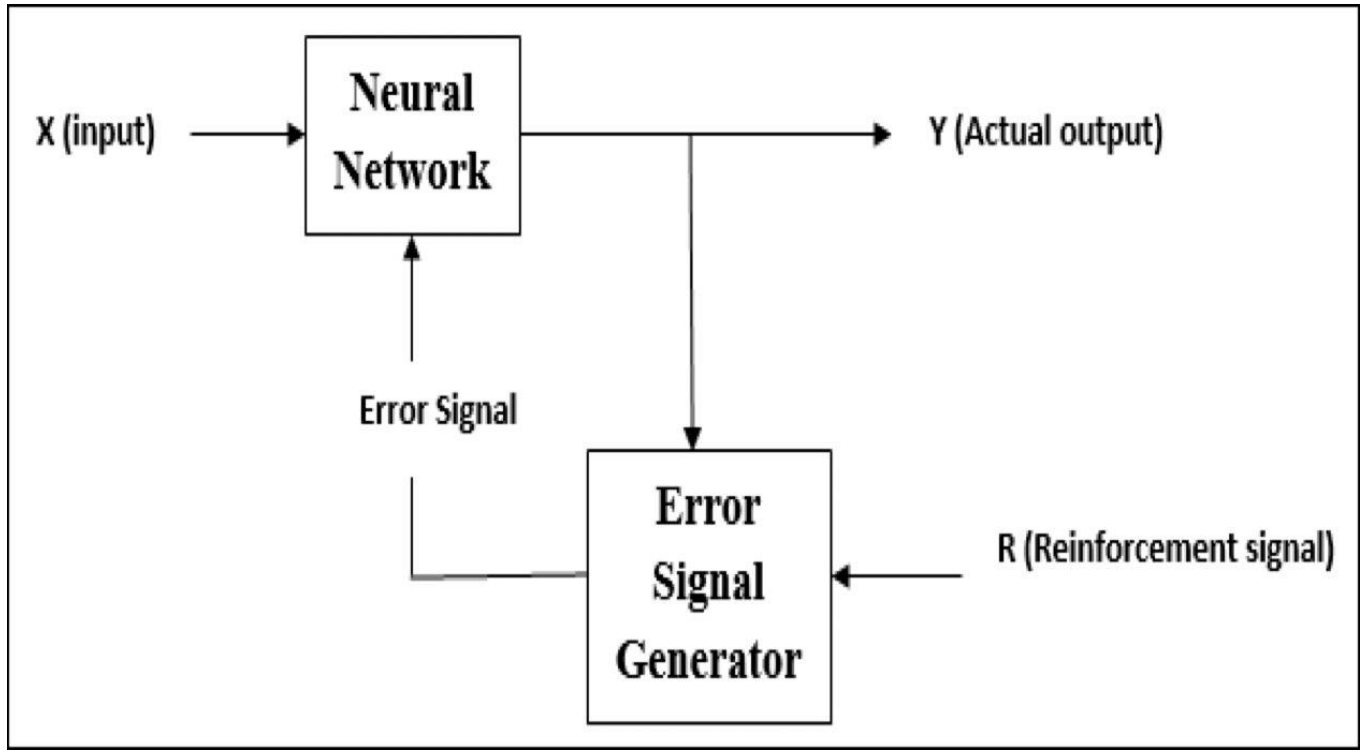
Unsupervised Learning



- The algorithms that analyze and cluster unlabeled dataset.
- There is no feedback from the environment as to what should be the desired output and if it is correct or incorrect. Hence, in this type of learning, the network itself must discover the patterns and features from the input data, and the relation for the input data over the output.
- Approaches: Clustering, association and dimensionality reduction

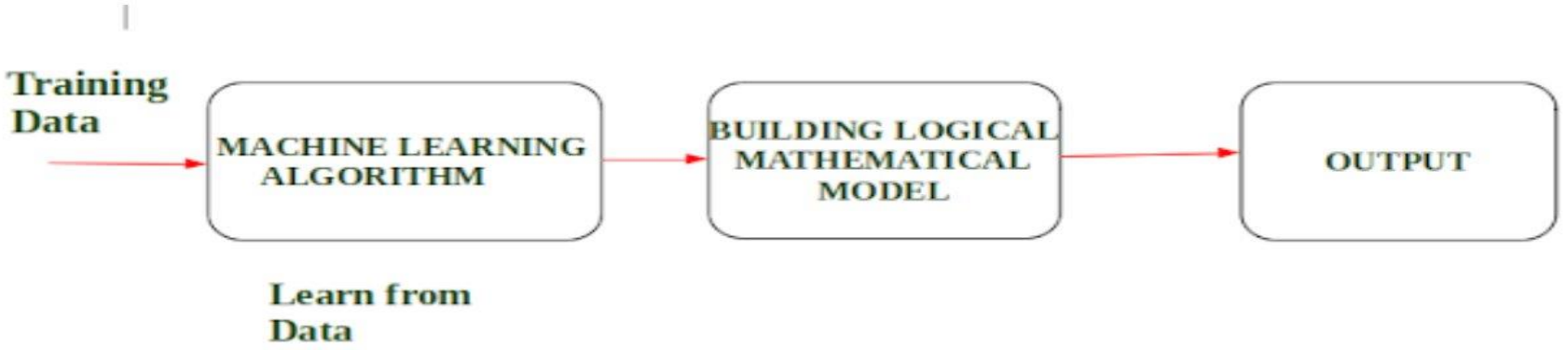
Reinforcement Learning(An action learner)

- process in which autonomous agents learn to make decisions by interacting with their environment.
- consists of the relationship between an agent, environment, and goal.
- Eg: robots, self-driving cars
- autonomous agents learn to perform a task by **trial and error** in the absence of any guidance from a human user.
- It particularly addresses sequential decision-making problems in uncertain environments,



Aspects of developing system

- **“Machine Learning enables a Machine to Automatically learn from Data, Improve performance from an Experience and predict things without explicitly programmed.”---Arthur Samuel.**
- critical aspects: data preparation (training and testing data), concept representation, and evaluation processes



Learning Systems

- A learning system in machine learning refers to a structured framework that enables machines to learn from data, recognize patterns, and make informed decisions with minimal human guidance.
- These systems are at the core of artificial intelligence, allowing models to adapt and improve as they process more data.

Components of learning systems

- **Data Input:** The raw or processed data supplied to the system forms the basis for model training. Both the **quality and the volume of this data** play a crucial role in determining the overall performance of the system.
- **Learning Algorithm:** This is the **central component that governs the learning process**. Algorithms such as Linear Regression, Neural Networks, and Decision Trees are designed to address specific problem types and to optimize the model's performance.
- **Model Output:** Once training is complete, the system produces outputs in the form of predictions, classifications, or recommendations derived from the input data. These outputs are progressively improved through repeated learning cycles. **Metrics to improve learning quality**
- **Feedback Mechanism:** This essential component supports continuous improvement by comparing predicted outcomes with actual results. **The feedback enables the system to adjust its parameters and minimize errors,** allowing the model to learn and adapt over time.

Steps in developing the system

- The development of a machine learning system is a **systematic and iterative process** that involves several well-defined stages.
- Steps involved in the development of a machine learning system are:

1. **Defining the problems and objectives**
2. **Data Collection and Preparation**
3. **Choosing the Training Experience**
4. **Selecting the Target Function**

5. Choosing a Representation for the Target Function

6. Selecting a Function Approximation Algorithm

7. Training the Model

8. Evaluating Model Performance

9. Iterative Refinement

1. **Defining the problems and objectives**

- The foundation of a learning system begins with a **clear problem definition**
- Identifying the type of task the system aims to solve, such as **classification** (e.g., predicting whether an email is spam), **regression** (e.g., forecasting sales figures), or **clustering** (e.g., customer segmentation).
- define measurable performance objectives using **metrics** like accuracy, precision, recall, orF1-score

2. Data Collection and Preparation

- Collecting and preprocessing high-quality, relevant data is critical for the system's success.
- data collection must focus not only on quantity but also on accuracy, consistency, and relevance
- Sources of data: databases, sensors, web sources, surveys and forms
- Data Gathering: Data can come from different sources
- Importance of data quality
- Data Preprocessing: The process of transforming raw data into a **clean and usable format** suitable for machine learning algorithms. Data cleaning techniques, handling missing values, normalization/encoding

3. Choosing the Training Experience

- Decide on the type of training experience based on the problem and the nature of the data:
- It refers to **deciding what data and experiences the learning algorithm will use to learn a task**.
- **Supervised Learning**: Best for tasks where labeled data is available, such as predicting loan defaults.
- **Unsupervised Learning**: Suitable for uncovering hidden patterns in unlabeled data, such as clustering customers.
- **Reinforcement Learning**: Ideal for dynamic environments like robotics or gaming, where the system learns through trial and error.

4. Selecting the Target Function

- The **target function** defines the relationship between inputs (features) and desired outputs(predictions).
- Clearly defining this function ensures the algorithm focuses on the right outcomes during training.
- Eg: loan approval----> reject/accept

5. Choosing a Representation for the Target Function(Concept Representation)

- Choosing the appropriate representation for the target function depends on the problem's complexity and data characteristics:
- **It determines how the system represents learned knowledge and how accurately it can generalize to unseen data.**
- **Decision Trees**: Effective for hierarchical decision-making tasks.
- **Neural Networks**: Suitable for handling non-linear relationships in large, complex datasets.
- **Linear Models**: Best for interpretable, straightforward problems.
- This step requires balancing model complexity with interpretability and computational efficiency.

6. Selecting a Function Approximation Algorithm

- A **function approximation algorithm** is a method used to **learn or estimate an unknown target function** by finding the best possible function (hypothesis) from a given hypothesis space that maps inputs to outputs.
- This ensures Accurate learning of input–output relationships
- The chosen algorithm :
- Searches the **hypothesis space** (set of possible functions)
- Adjusts model parameters
- Minimizes error or maximizes performance
- Algorithms: Gradient descent, Backpropagation, Least Square Method...

7. Training the Model

- Train the chosen model using the prepared data
- Iteratively feeding the data into the algorithm.
- Adjusting model parameters to minimize errors using techniques like backpropagation in neural networks.
- Monitoring the training process to avoid issues like overfitting, which occurs when the model performs well on training data but poorly on unseen data.

8. Evaluating Model Performance

- Evaluating the model ensures it generalizes well to new data
- Validation is the process of assessing the model's performance during development and tuning
- Common evaluation methods include:
  - Hold-out validation
  - k-fold cross-validation
  - Performance metrics such as accuracy, precision, recall, F1-score, and mean squared error
  - Evaluation helps in identifying issues such as overfitting or underfitting and supports model refinement and optimization.

9. **Iterative Refinement**

- a trial-and-improvement cycle.
- Iterative refinement helps improve the model's accuracy, robustness, and generalization.